

IoT BASED ELECTROCARDIOGRAM AND BEATS PER MINUTE AND ALERT SYSTEM

A Project Report submitted in Partial Fulfillment of the requirements for the award of the degree of Bachelor of Technology in

Instrumentation Engineering

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Submitted by

Manash Pratim Nath	(220710015017)
Nirjan Hazarika	(220710015019)
Priyanka Dutta	(220710015021)
Sagar Kalita	(220710015027)

Under the Supervision of

Mr. Kalyan Bhattacharjee

Assistant Professor

Instrumentation Engineering

Jorhat Engineering College

DEPARTMENT OF INSTRUMENTATION ENGINEERING

JORHAT ENGINEERING COLLEGE

PIN-785007

DECLARATION BY THE CANDIDATES

We hereby declare that this B. Tech project report titled "IoT-Based ECG and BPM Monitoring and Alert System " is an authentic record of our own work carried out under the supervision of **Mr Kalyan Bhattacharjee**, Assistant Professor, Department of Instrumentation Engineering, Jorhat Engineering College.

We affirm that the work presented in this report has not been submitted by us for the award of any degree, diploma, fellowship, or other similar titles at any other university or institution. All sources of information and help have been duly acknowledged.

Roll No.	Name of the candidate	Signature
220710015017	Manash Pratim Nath	
220710015019	Nirjan Hazarika	
220710015021	Priyanka Dutta	
220710015027	Sagar Kalita	

Signature of the Supervisor:

Name:

Date:

Signature of the HOD:

Name:

Date:

CERTIFICATE FROM EXAMINER

This is to certify that the project report entitled, “**IoT Based ECG and BPM Monitoring and Alert System**” submitted by Manash Pratim Nath (220710015017), Nirjan Hazarika (220710015019), Priyanka Dutta (220710015021), Sagar Kalita (220710015027) in partial fulfilment of the requirements for the degree of **Bachelor of Technology** in Instrumentation Engineering under ASTU has been examined by me and is found satisfactory for the award of the degree.

This approval does not necessarily endorse or accept every statement made, opinion expressed, or conclusion drawn as recorded in the report. It only signifies the acceptance of this dissertation for the purpose for which it is submitted.

Date:

(External Examiner)

Place: Jorhat

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Signature:

Date:

Place: Jorhat Engineering College, Jorhat, Assam

ABSTRACT

The rapid growth of the Internet of Things (IoT) has changed modern healthcare systems by allowing remote, real-time, and continuous tracking of physiological parameters. Cardiovascular diseases (CVDs) are still among the top causes of death worldwide. This emphasizes the need for early detection and ongoing cardiac monitoring. Traditional monitoring systems often depend on expensive medical equipment, frequent hospital visits, and constant supervision by healthcare professionals. This makes them harder to access for long-term monitoring and emergency detection.

To overcome these challenges, this project introduces a low-cost, IoT-based real-time heart monitoring system that uses an AD8232 ECG sensor, a MAX30102 sensor, and an ESP32 microcontroller. The AD8232 sensor captures the heart's electrical activity as an Electrocardiogram (ECG) signal and calculates the heart rate in beats per minute (ECG BPM). At the same time, the MAX30102 sensor provides its own heart rate measurement. The ESP32 serves as the main processing unit, gathering data from both sensors and conducting real-time analysis.

The proposed system continuously compares the ECG BPM from the AD8232 sensor with the heart rate from the MAX30102 sensor. A large difference between these values could signal abnormal physiological conditions, leading the system to trigger an alert. In these cases, a buzzer sounds and an alert notification is sent through Telegram to ensure timely awareness and response.

For data visualization and monitoring, the ESP32 runs a local web server that shows the ECG waveform, ECG BPM, heart rate values, and system status in real time. Users can access the monitoring interface from any device connected to the same network, making observation convenient without needing special software.

The system offers a dependable and effective solution for real-time cardiac monitoring by blending ECG signal capture, heart rate analysis, sensor data comparison, web-based visualization, and automatic alert creation. This approach improves monitoring accuracy and aids early detection of abnormal heart conditions through ongoing and accessible health monitoring.

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CHAPTER 1: INTRODUCTION

1.1 INTRODUCTION

Monitoring heart rate is crucial for assessing cardiovascular health and spotting issues early. Continuous tracking of heart activity offers important insights about an individual's condition and helps with timely interventions in abnormal situations. Recent advances in embedded systems and wireless communication technology have led to efficient and secure real-time monitoring systems.

This project presents a secure heart rate monitoring system that uses the ESP32 microcontroller, the AD8232 ECG sensor, and the MAX30102 sensor. The AD8232 sensor captures the heart's electrical activity as an ECG waveform. Besides recording ECG signals, the data is processed to determine the heart rate in beats per minute (ECG BPM). The MAX30102 sensor is also used to get heart rate values independently. The heart rate readings from both sensors are continuously analysed and compared to improve the monitoring system's reliability and accuracy.

The ESP32 acts as the central processing unit for the system. It gathers sensor data, processes the ECG waveform, calculates the ECG BPM, receives heart rate measurements from the MAX30102 sensor, and conducts comparison analysis. If a significant difference appears between the ECG BPM and the heart rate from the MAX30102 sensor, the system marks it as a possible abnormal condition and sends an alert notification.

The system also hosts a local web server on the ESP32, which allows users to view the ECG signal, ECG BPM, heart rate values, and system status in real time through a web browser on the same network. This makes monitoring convenient without the need for special software.

The main goal of this project is to create a secure and efficient heart monitoring system. It will acquire ECG signals, calculate ECG BPM, compare heart rate values from multiple sensors, generate alerts during abnormal conditions, and provide real-time data visualization through a local host interface.

1.2 PROBLEM STATEMENT

Cardiovascular diseases are among the leading causes of death worldwide. Continuous monitoring of heart activity is essential for early detection and prevention of serious health issues. Traditional cardiac monitoring systems typically rely on hospital equipment and periodic medical exams, which do not provide constant observation of a patient's heart condition. Because of this, abnormal cardiac events can go undetected between monitoring sessions.

Heart rate monitoring systems that use a single sensing method may suffer from signal noise, sensor placement errors, motion artifacts, or environmental disturbances, leading to inaccurate measurements. Such inaccuracies can lower the reliability of the monitoring system, potentially causing false alarms or missed detection of abnormal conditions.

Therefore, there is a need for a reliable real-time monitoring system that can acquire ECG signals, calculate ECG BPM, and confirm heart rate measurements using an additional sensing source. The system should also offer continuous monitoring, real-time visualization, and immediate alerts when abnormal conditions are detected.

This project aims to tackle these challenges by developing an ESP32-based heart monitoring system. It will acquire ECG signals using the AD8232 sensor, calculate ECG BPM, compare it with the heart rate measured by the MAX30102 sensor, and generate alerts when significant deviations occur. The monitored data will be displayed through a local host web server to enable real-time observation and improve the reliability of heart rate monitoring.

1.3 OBJECTIVE OF THE PROJECT

The primary objective of this project is to develop a real-time heart monitoring system capable of acquiring, processing, and analyzing cardiac signals for reliable health monitoring. The specific objectives of the project are as follows:

- a) To acquire the electrical activity of the heart using the AD8232 ECG sensor and display the ECG waveform in real time.

- b) To calculate the heart rate in Beats Per Minute (ECG BPM) from the acquired ECG signal.
- c) To measure heart rate using the MAX30102 sensor and obtain an independent BPM value.
- d) To compare the ECG BPM obtained from the AD8232 sensor with the BPM measured by the MAX30102 sensor for improved monitoring reliability.
- e) To detect significant deviations between the two heart rate measurements and identify potential abnormal conditions.
- f) To generate an alert through a buzzer whenever abnormal conditions are detected.
- g) To develop a local host web server using the ESP32 for real-time visualization of ECG signals, ECG BPM, sensor BPM values, and system status.
- h) To design and implement a compact hardware system using the ESP32 microcontroller for efficient data acquisition and processing.
- i) To provide a reliable, low-cost, and user-friendly solution for continuous heart monitoring and early abnormality detection.

1.4 SCOPE AND SIGNIFICANCE

Scope of the Project

The scope of this project is to develop a real-time heart monitoring system capable of acquiring ECG signals and heart rate data using the AD8232 ECG sensor and MAX30102 sensor. The system processes the acquired signals using the ESP32 microcontroller and compares the ECG BPM with the BPM obtained from the MAX30102 sensor. The monitored data is displayed through a local host web server, allowing users to observe heart activity in real time. In addition, the system generates alerts through a buzzer whenever abnormal conditions are detected.

- a) To acquire ECG signals using the AD8232 ECG sensor.

- b) To calculate ECG BPM from the acquired ECG waveform.
- c) To measure heart rate using the MAX30102 sensor.
- d) To compare ECG BPM with the BPM obtained from the MAX30102 sensor.
- e) To detect abnormal conditions based on significant BPM variations.
- f) To display ECG waveform and heart rate data on a local host web page.
- g) To generate alerts through a buzzer when abnormalities are detected.
- h) To provide real-time monitoring using the ESP32 microcontroller.
- i) To develop a compact and portable heart monitoring system.

Significance of the Project

Continuous monitoring of heart activity is important for the early detection of cardiovascular abnormalities and timely medical attention. The proposed system improves monitoring reliability by comparing heart rate values obtained from two different sensors. Real-time visualization and automatic alert generation enhance the effectiveness of the monitoring process and provide immediate notification during abnormal conditions.

- a) Provides continuous real-time monitoring of heart activity.
- b) Assists in the early detection of abnormal cardiac conditions.
- c) Improves reliability through dual-sensor heart rate comparison.
- d) Enables quick response through buzzer .
- e) Allows convenient monitoring through a local host web interface.
- f) Reduces dependence on continuous hospital-based monitoring.
- g) Utilizes low-cost and easily available hardware components.
- h) Demonstrates the application of IoT and embedded systems in healthcare.
- i) Enhances patient safety through automated abnormality detection.
- j) Provides a foundation for future advanced healthcare monitoring systems.

CHAPTER 2: LITERATURE REVIEW

2.1 LITERATURE REVIEW

1. Going through the paper [1] the following points could be observed.
 - a. Cardiovascular diseases are the leading global killer (~17.9M deaths/year).
 - b. Lack of cardiologists and expensive ECG machines limit care access.
 - c. Create a low-cost, IOT-enabled ECG system for remote monitoring and real-time alerts.
 - d. AD8232 Sensor, ESP32 Module, Ubidots Cloud, Arduino IDE.
 - e. ECG signals are processed by AD8232, sent via ESP32 and displayed on the Ubidots cloud for doctor alerts.
 - f. Portable, affordable, and ideal for telemedicine and rural healthcare.
2. Going through paper [2] the following point could be observed:
 - a) Early detection of medical issues and emergency alerts.
 - b) MAX30100 (heart rate & SpO2), DHT11 (temperature), NodeMCU, and Cloud Server.
 - c) Sensor data is AES-encrypted by NodeMCU, sent to the cloud, and alerts doctors on abnormal readings.
 - d) Three layers: Patient, Web Server, and Hospital for monitoring and data management.
 - e) Portable, low-cost, secure, supports telemedicine and remote care.
3. Going through paper [3], the following points are observed

- a) To provide continuous, remote monitoring of patients' vital signs - especially for elderly chronically ill people - using affordable IoT technology.
 - b) To design a low-cost, wearable system that measures and transmits vital health data (heart rate, temperature, and SpO₂) to doctors or caregivers in real time.
 - c) Sensors: Heart rate, temperature, SpO₂, accelerometer
 - d) Microcontroller: Arduino / NodeMCU (ESP8266)
 - e) Cloud Platform: Blynk
 - f) Communication: Wi-Fi
 - g) Power: Rechargeable battery
 - h) Health data is collected by sensors, processed by the microcontroller, transmitted via Wi-Fi to the cloud for real-time display, and alerts are generated for abnormal values.
 - i) Real-time remote health monitoring
 - j) Portable and cost-effective
4. Going through paper [4], the following points are observed:
- a) To enable real-time, remote monitoring of vital signs and reduce hospital visits through IoT technology.
 - b) To design a smart system that measures and transmits health data (heart rate, SpO₂, temperature) to the cloud and alerts doctors or family in emergencies.
 - c) Smartwatch/sensors, microcontroller (ESP32/Arduino), smartphone, ThingSpeak cloud, IFTTT, Wi-Fi.
 - d) Sensors collect body data, the microcontroller sends it to ThingSpeak via Wi-Fi for online viewing, and IFTTT triggers alerts when abnormal values are detected.
 - e) Real-time monitoring, remote access, quick alerts, reduced hospital visits, and data storage for analysis.
5. Going through paper [5], the following points are observed:
- a) To develop a wearable IOT system (WISE) for real-time remote health monitoring without smartphone dependency.
 - b) To use sensors for heart rate, temperature and BP that send data to the cloud for analysis and alerts.

- c) Heartbeat, LM35, BP sensors, Arduino, RFID, WIFI cloud, web GUI.
- d) Sensors collect data, Arduino sends it via Wi-Fi to the cloud for storage and display, and alerts are sent for emergency response.
- e) Real-time monitoring, cloud data access, instant alerts, and stored health records.

2.2 RESEARCH GAP

Several systems for real-time health monitoring based on heart rate, using an ECG sensor, a pulse sensor, and IoT, have been developed. The majority of systems have the sole function of determining heart rate using a single sensor, which has several problems, such as inaccurate readings due to noise, errors in sensor positioning, motion artefacts, and interference. A great number of studies present either a heart rate monitor or ECG signal acquisition individually, but no direct comparison between ECG-based BPM and other sensors' BPM is made. Hence, the validation of measured values is often limited. A large number of existing systems provide local visualisation of data and have no system for real-time alarm generation to alert the user to abnormal conditions. Lastly, many of these systems utilise external software or a cloud-based environment, thus making them complicated to implement and reliant on outside services. This system intends to create an easy-to-use, low-cost and highly reliable heart monitoring system that provides real-time monitoring, validation of heart rate values and instant alert generation.

Research Gaps identified:

- a) Most systems are based on data from only a single sensor for heart rate determination.
- b) Comparison of ECG-based BPM with BPM obtained by another, independent sensor has only been reported by a few systems.
- c) Lack of a heart rate validation method that enhances reliability.
- d) Lack of real-time detection of abnormalities in a large number of systems.
- e) Lack of automated alerts in several heart rate monitoring systems.
- f) The existing solutions often depend on external software and a cloud-based environment.
- g) Existing low-cost and user-friendly systems are not readily available.
- h) Integration of ECG visualisation, BPM comparison and alerts is often not provided in a single platform.

CHAPTER 3: THEORITICAL BACKGROUND

3.1 INTERNET OF THINGS (IoT)

The **Internet of Things (IoT)** refers to a network of physical devices that are connected to the internet to collect, share, and exchange data automatically. These devices include sensors, microcontrollers, appliances, vehicles, and machines that are equipped with electronics, software, and communication technologies. IoT allows these devices to work together intelligently without human intervention.

In an IoT system, sensors collect data from the environment or the human body, which is then processed by a microcontroller or processor. The processed data is transmitted through the internet to a cloud platform, where it can be stored, analysed, and visualised. Users can access this information using smartphones, computers, or other connected devices from anywhere in the world.

IoT is widely used in various fields such as smart homes, agriculture, industry, transportation, and healthcare. In healthcare, IoT enables continuous monitoring of vital signs like ECG, blood pressure, heart rate, and oxygen level, making early diagnosis and remote treatment possible.

Overall, IoT improves efficiency, saves time, and enhances the quality of life by making systems smarter, faster, and more connected. It plays a key role in developing modern smart healthcare systems like the one proposed in this project.[6]

3.2 ELECTROCARDIOGRAM (ECG)

An **Electrocardiogram (ECG)** is a medical test that records the electrical activity of the heart over a period of time. The heart produces small electrical signals each time it beats, and these signals control the contraction and relaxation of the heart muscles. By measuring these signals, an ECG provides important information about the heart's rhythm, rate, and overall condition.

An ECG sensor detects these electrical impulses using electrodes placed on the skin, usually on the chest, arms, or legs. The signals are then converted into electrical waveforms that can be displayed on a screen or recorded for analysis. These waveforms show different parts of

the heartbeat, such as the P wave, QRS complex, and T wave, which help doctors understand how the heart is functioning.

ECG is widely used to detect various heart problems such as arrhythmia (irregular heartbeat), heart attack, and other cardiac abnormalities. In this project, the ECG sensor is connected to the ESP32 microcontroller, which captures and transmits the ECG data to an IoT cloud platform for real-time monitoring.

This IoT-based ECG monitoring system helps in early detection of heart problems, provides continuous observation, and allows doctors to access patient data remotely.[7]

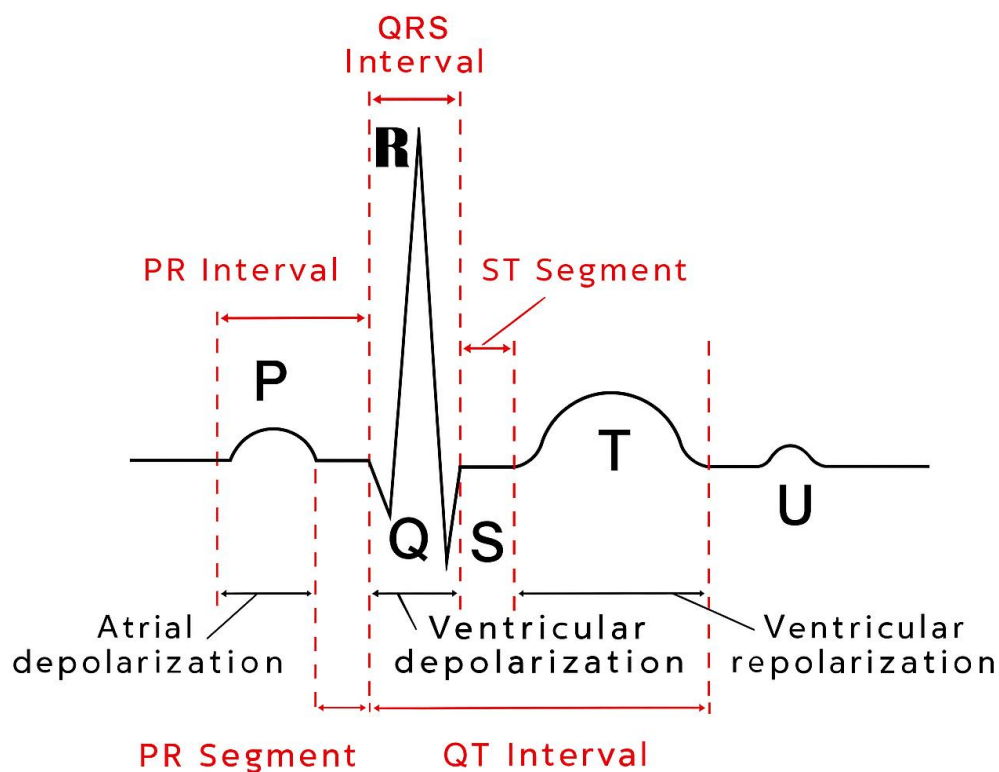


FIG: ECG WAVEFORM

3.3 BEATS PER MINUTE(BPM)

Beats Per Minute (BPM) refers to the number of times the heart beats in one minute and is a basic indicator of heart activity. It is widely used to quickly understand how the heart is functioning under different conditions such as rest, exercise, or stress. For most adults, a normal resting BPM typically lies between 60 and 100, although it can vary depending on individual health and fitness levels.

Measuring BPM is important because it helps in identifying unusual heart behavior. A higher BPM may indicate physical exertion, anxiety, or possible health issues, while a lower BPM is often seen in well-trained individuals but may also require attention in some cases. Regular monitoring can therefore help in early detection of abnormalities.

In this project, BPM is measured using the MAX30102 Pulse Oximeter Sensor, which works by sensing changes in blood flow using light. The data is then processed by the ESP32 Dev Board and displayed for real-time monitoring.

Overall, BPM is a simple yet effective parameter for tracking heart health and plays a key role in modern health monitoring systems.

3.4 ESP32

The **ESP32** is a low-cost, low-power, and high-performance microcontroller developed by Espressif Systems. It is widely used in Internet of Things (IoT) applications because of its built-in Wi-Fi and Bluetooth capabilities, which make it ideal for wireless communication and real-time data transmission. The ESP32 is an upgraded version of the earlier ESP8266 module and offers more processing power, memory, and versatility.

The ESP32 features a dual-core processor, multiple GPIO (General Purpose Input/Output) pins, Analog-to-Digital Converters (ADC), Digital-to-Analog Converters (DAC), and support for various sensors and peripheral devices. It can easily interface with biomedical sensors such as ECG and BP sensors to collect, process, and transmit patient data to an IoT cloud platform.

In this project, the ESP32 acts as the central processing and communication unit. It receives analog signals from the ECG and BP sensors, converts them into digital data, and then uploads the data wirelessly to an IoT cloud service such as ThingSpeak or Blynk for visualization and remote monitoring.

The main advantages of using the ESP32 include low cost, small size, low power consumption, and reliable wireless connectivity, making it an excellent choice for portable healthcare and IoT-based medical applications.[9]

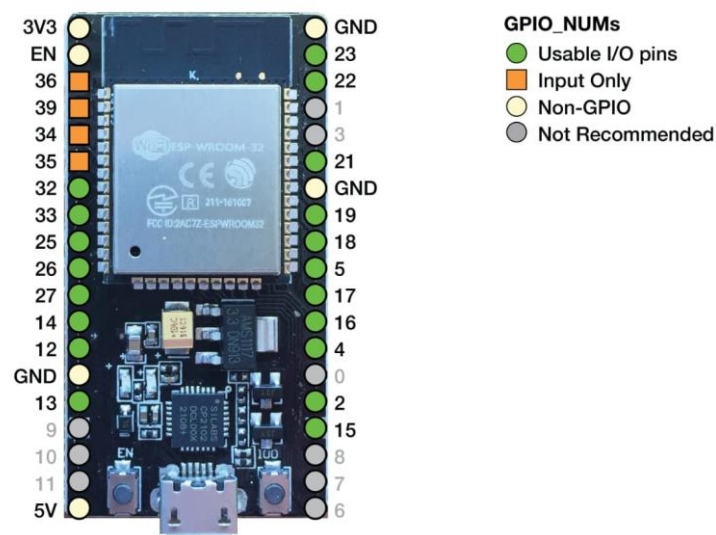


FIG: ESP 32 PIN DIAGRAM

3.5 ECG SENSOR (AD8232)

The **AD8232** is a low-power, integrated signal conditioning block designed for measuring the electrical activity of the heart. It is widely used in portable ECG and heart rate monitoring applications. The sensor amplifies and filters the small biopotential signals generated by the heart, providing a clear and stable ECG waveform. It uses three electrodes — typically connected to the left arm, right arm, and right leg — to detect heart activity. In this project, the AD8232 sensor is connected to the ESP32 microcontroller, which processes and transmits the ECG data to an IoT cloud platform for real-time monitoring.[10]

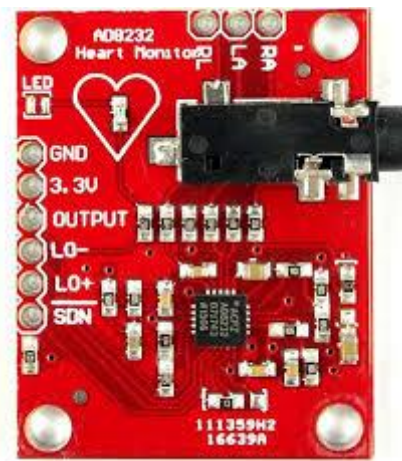


FIG: AD8232 SENSOR

3.6 MAX30102 (PULSE OXIMETER SENSOR)

The MAX30102 Pulse Oximeter Sensor is a small and efficient sensor used to measure heart rate (BPM) and blood oxygen levels (SpO₂). It works using red and infrared light to detect changes in blood flow through the skin.

This sensor communicates easily with microcontrollers like the ESP32 Dev Board using the I2C protocol and operates on low power, making it suitable for portable health monitoring systems.

In this project, it is used to provide real-time heart rate data, helping in continuous monitoring of the user's health.

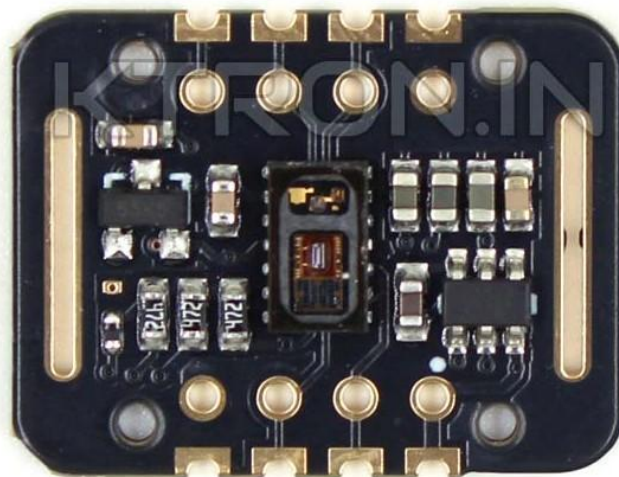


FIG : MAX30102

CHAPTER 4: METHODOLOGY

4.1 SYSTEM OVERVIEW AND DESIGN APPROACH

The system presented here is a real-time heart monitoring system used for collecting, processing, analysing, and displaying cardiac data using an ESP32 microcontroller, an AD8232 ECG sensor, and a MAX30102 sensor. The AD8232 sensor measures the electrical signals produced by the heart and produces an ECG waveform, which provides the ECG BPM of the user, while the MAX30102 sensor measures the heart rate separately.

ESP32 acts as the main unit of the system by collecting the data from both the sensors, and the real-time analysis is done in it. The ECG BPM from the AD8232 is compared with the heart rate given by the MAX30102. If there is an abnormal difference between the two, a condition is detected, and the alert system is triggered.

The ECG waveform, ECG BPM, MAX30102 BPM and the system status are displayed through the ESP32 with a local host web server, and it also gives an audible alert through the buzzer.

Main components of the system:

- a. ESP32 microcontroller
- b. AD8232 ECG sensor
- c. MAX30102 Heart Rate sensor
- d. Buzzer alert unit
- e. Local host web server
- f. Power supply unit

Design approach

The proposed system has a design approach that follows the following steps:

Step 1: Signal Acquisition

- The AD8232 sensor takes the ECG signal.
- Heart rate is measured by MAX30102 separately.
- Sensor data is sent to the ESP32.

Step 2: Data processing

- ECG waveform is processed in the ESP32.
- ECG BPM is extracted from the received signal.
- The BPM measured by MAX30102 is retrieved.

Step 3: Data comparison

- ECG BPM is compared with the heart rate given by the MAX30102.
- The difference is continuously observed.

Step 4: Abnormality detection

- When the difference between the two values exceeds a preset limit, an abnormality is detected.
- The alerting system is triggered.

Step 5: Data Visualisation

- The ECG waveform is viewed in the local host web page.
- The real-time ECG and heart rate are updated, and the status is shown.

Step 6: Alerting

- Buzzer is turned ON.

Step 7: Real-time monitoring

- The system continues the above-mentioned operations in a real-time loop.

4.2 BLOCK DIAGRAM OF THE SYSTEM

This is the working principle of the proposed system, which is to measure the activity of the heart in real time using the ECG sensor AD8232, the MAX30102 sensor and the ESP32 microcontroller. The AD8232 sensor measures the ECG waveform and ECG BPM, whereas the heart rate is measured using the MAX30102 sensor, and the BPM measured from the sensor is also calculated. The ECG waveform, ECG BPM, MAX30102 BPM and system status are displayed on the local host web server. The BPM of ECG and heart rate is compared, and when

a difference occurs between these values, it indicates that there is an anomaly in the measured heart rate value, and a buzzer sound is produced.

Function Flow of Block Diagram.

- ECG electrodes are placed to acquire heart signals.
- The AD8232 sensor reads the ECG signal and gives the ECG waveform and ECG BPM.
- The MAX30102 sensor reads heart rate, which results in BPM.
- ESP32 receives sensor data and reads.
- ECG BPM and MAX30102 BPM comparison happens at the BPM comparison unit.
- If the bpm of ECG and heart rate are within the desired threshold, then the system is working fine; otherwise, it detects the abnormal condition, and then the buzzer sound will start.
- The ECG waveform, ECG BPM, MAX30102 BPM and system status are displayed on the local host web server.
- If there is a disparity in the measured BPM value, then the system determines the anomaly.
- The buzz sound is played.
- Steps are again iterated for real-time heart activity monitoring.

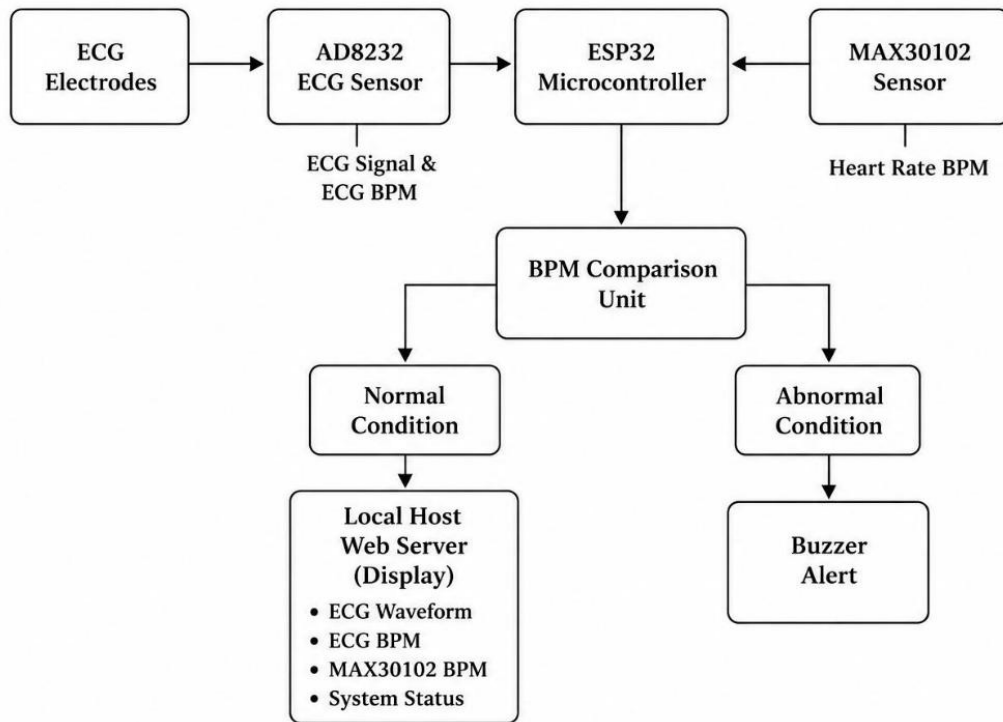
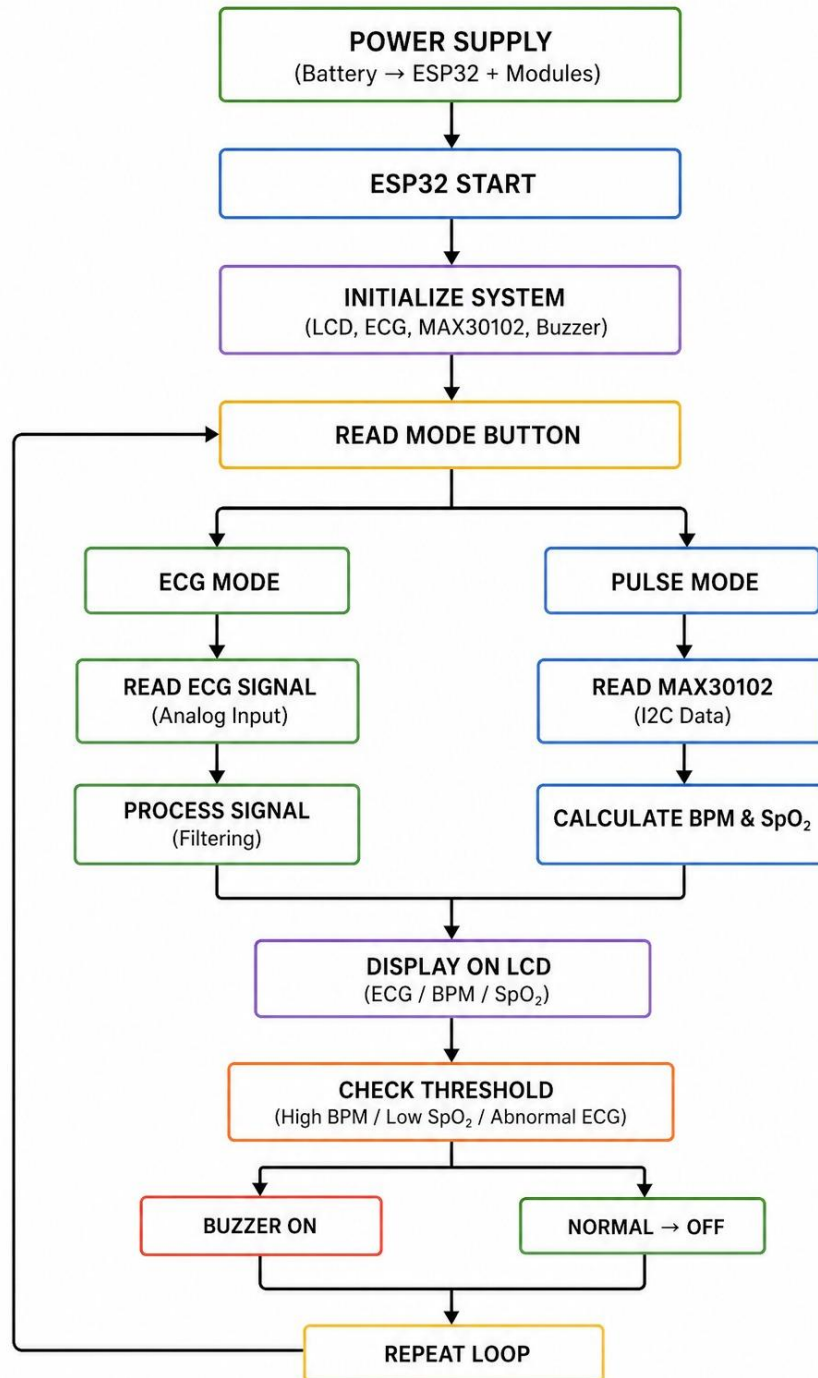


FIG: BLOCK DIAGRAM OF THE SYSTEM

4.2 FLOW CHART



4.3 ELECTRICAL CIRCUIT DIAGRAM

Pin connection table

1. MAX30102 → ESP32

MAX30102 Pin	ESP32 Pin
VIN	3.3V
GND	GND
SDA	GPIO 21
SCL	GPIO 22
INT	Not Connected

2. AD8232 ECG Sensor → ESP32

AD8232 Pin	ESP32 Pin
3.3V	3.3V
GND	GND
OUTPUT	GPIO 34 (ADC)
LO+	GPIO 32
LO-	GPIO 33
SDN	Not Connected

3. I2C LCD (16×2) → ESP32

LCD I2C Pin	ESP32 Pin
VCC	5V (VIN)
GND	GND
SDA	GPIO 21
SCL	GPIO 22

4. Buzzer → ESP32

Buzzer Pin	ESP32 Pin
Positive (+)	GPIO 25
Negative (−)	GND

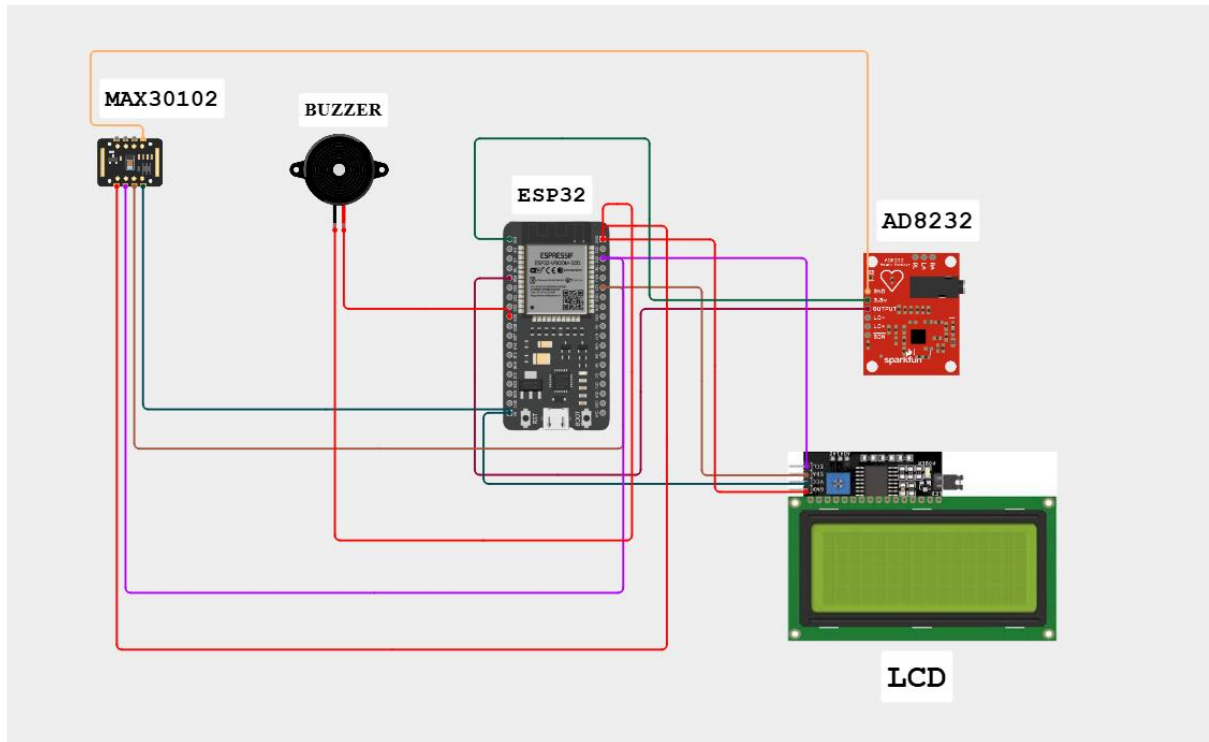


FIG: MODEL CIRCUIT DIAGRAM

4.4 ECG BPM CALCULATION

The AD8232 ECG sensor acquires the electrical activity of the heart and generates an ECG waveform. The acquired signal contains various components such as the P-wave, QRS complex, and T-wave. Among these components, the R-peak of the QRS complex is the most prominent feature and is used for heart rate calculation.

The ESP32 continuously monitors the ECG signal and detects consecutive R-peaks. The time interval between two successive R-peaks, known as the RR interval, is measured. The heart rate in Beats Per Minute (BPM) is then calculated using the RR interval. Since one minute contains 60 seconds, the BPM can be determined using the following equation:

The heart rate is calculated by measuring the time interval between two consecutive R-peaks of the ECG waveform, known as the RR interval.

$$\text{RR Interval} = R_2 - R_1$$

where,

R_1 = Time of the first R-peak

R_2 = Time of the second R-peak

The heart rate in Beats Per Minute (BPM) is calculated using the following equation:

$$\text{BPM} = 60 / \text{RR Interval}$$

where the RR interval is measured in seconds.

If the RR interval is measured in milliseconds, the BPM can be calculated as:

$$\text{BPM} = 60000 / \text{RR Interval}$$

4.5 LOCAL HOST WEBPAGE

A local host webpage is designed with an ESP32 microcontroller, which is capable of monitoring real-time cardiac activity of the patient. ESP32 is also capable of being a web server; it will host a webpage. So the user just needs to have any devices on the same WiFi network, and they will be able to see the webpage of monitoring on their browsers. There is no software required to see this webpage.

The webpage shows the ECG waveform on the browser, which is collected by the AD8232 sensor. The waveform is constantly updated. The webpage also shows the ECG BPM, which is derived from the ECG waveform and the MAX30102 BPM. These values are also updated from time to time.

The webpage also shows the system status by comparing the ECG BPM and MAX30102 BPM. It shows normal status normally. If the difference between ECG BPM and MAX30102 BPM is big enough, it would show an abnormal status and activate the buzzer.

The local host webpage gives the monitoring of the ECG signal on the same WiFi with a good user experience.

CHAPTER 5: SIMULATION AND RESULT

CHAPTER 6: CONCLUSION

6.1 SUMMARY OF WORK

The work carried out in this project includes design and development of the real-time heart monitoring system with the help of the ESP32 microcontroller, the AD8232 ECG sensor and the MAX30102 sensor. The AD8232 sensor is used for collecting the ECG waveform and calculating the ECG heart rate in Beats Per Minute(ECG BPM), while the MAX30102 sensor is used for finding the individual heart rate. The values from two sensors are compared constantly after getting the results from both sensors and processed by the ESP32 to improve the monitoring reliability. The local host web server is being used by ESP32, which allows for monitoring the ECG waveform, ECG BPM, MAX30102 BPM and system status. Through the web browser, it can be viewed easily once it is connected to the same network. Safety of the system is achieved using a buzzer that alerts if the difference between the heart rates calculated from two sensors exceeds a certain limit. The hardware designs involve sensor interfacing, signal collection, signal processing, web-based monitoring and alert generation. PCB design and hardware implemented in this project will result in a low-power consumption monitoring device. The system shows the combination of biomedical sensors, embedded systems and web-based technology. The overall system will prove to be a low-cost and reliable system that monitors the user's heart rate in real time and also displays if there is any abnormal condition with the help of a buzzer.

6.2 KEY OUTCOMES AND OBSERVATIONS

Key Outcomes

- Successfully acquired ECG signals using the AD8232 sensor.
- Calculated ECG BPM from the ECG waveform.
- Measured heart rate using the MAX30102 sensor.
- Compared ECG BPM with MAX30102 BPM in real time.
- Developed a local host webpage for monitoring.
- Implemented a buzzer alert for abnormal conditions.

Observations

- The system provided stable real-time monitoring.

- ECG BPM and MAX30102 BPM showed similar values under normal conditions.
- Minor variations were observed due to sensor characteristics.
- The local host webpage displayed data accurately.
- The alert system responded effectively when abnormal conditions were detected.

6.3 LIMITATIONS OF THE CURRENT DESIGN

- Proper placement of ECG electrodes is required for accurate readings.
- Body movement can affect the ECG signal and reduce accuracy.
- The system works only within the same Wi-Fi network through the local host webpage.
- Small differences may occur between ECG BPM and MAX30102 BPM readings.
- The system monitors only heart activity and does not measure other health parameters.
- Continuous operation requires a stable power supply.
- Sensor readings may be affected by poor skin contact or external noise.
- The prototype is designed for monitoring purposes and not for medical diagnosis.

6.4 RECOMMENDATIONS FOR IMPROVEMENT

- Improve ECG signal accuracy and noise filtering.
- Develop a compact wearable design.
- Add cloud-based data storage and monitoring.
- Enhance BPM comparison accuracy.
- Improve the user interface of the monitoring webpage.
- Extend battery life and power management.

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